

Complete MCQ Set – Introduction to Robotics

1. Which of the following best defines *robotics*?
 - a) Study of artificial intelligence only
 - b) Engineering field combining mechanics, electronics, and computing (Ans.)
 - c) Branch of physics that studies motion
 - d) Subset of chemistry that studies automation

2. Which of the following robots operates **without direct human control** but within water environments?
 - a) UAV
 - b) UGV
 - c) AUV (Ans.)
 - d) RC Car

3. The **main drawback** of the hierarchical paradigm is:
 - a) Lack of real-time responsiveness (Ans.)
 - b) Too much reactive behavior
 - c) Poor mechanical design
 - d) Low computational demand

4. Which of the following correctly matches the “Sense–Plan–Act” cycle?
 - a) Motor → Brain → Sensor
 - b) Perception → Decision → Motion (Ans.)
 - c) Actuation → Control → Vision
 - d) Output → Input → Plan

5. Which law of robotics emphasizes obedience to human orders?
 - a) First Law
 - b) Second Law (Ans.)
 - c) Third Law
 - d) Fourth Law

6. A robot that maps an unknown area and updates its position is mainly dealing with:
 - a) Perception
 - b) Localization (Ans.)

- c) Planning
- d) Control

7. In a **reactive robotic system**, decisions are made:

- a) Based on global models
- b) Using pre-planned maps
- c) Using real-time sensor inputs only (Ans.)
- d) From human intervention

8. **Stepper motors** are primarily used in robots because they:

- a) Operate using compressed air
- b) Offer precise angular control without feedback (Ans.)
- c) Require complex hydraulic systems
- d) Run only on AC current

9. Which of the following components is considered part of the **control hardware** of a robot?

- a) PID algorithm
- b) Arduino board (Ans.)
- c) Localization code
- d) Trajectory planner

10. The **Four D's of Robotics** are used to identify:

- a) Dangerous materials
- b) Suitable robotic applications (Ans.)
- c) Human labor laws
- d) Robot sensor accuracy

11. Which of the following actuators uses pressurized fluid to generate motion?

- a) Pneumatic
- b) Hydraulic (Ans.)
- c) Electric
- d) Magnetic

12. Which subsystem is responsible for converting planned commands into physical motion?

- a) Sensing
- b) Planning
- c) Acting (Ans.)
- d) Power

13. In a robotic control system, PID control is primarily used for: (Out of syllabus just to a little understanding of students)

- a) Vision recognition
- b) Motion stabilization (Ans.)
- c) Data transmission
- d) Object classification

14. The key advantage of a hybrid deliberative/reactive architecture is:

- a) Complete elimination of sensors
- b) Fast decision-making with strategic planning (Ans.)
- c) Lack of modularity
- d) Elimination of global modeling

15. The term 'Localization' in robotics refers to:

- a) Finding optimal path to destination
- b) Identifying the robot's position in its environment (Ans.)
- c) Detecting obstacles using infrared sensors
- d) Communicating with nearby robots